

lab_05

Alexis Saldivar (PID: A69038083)

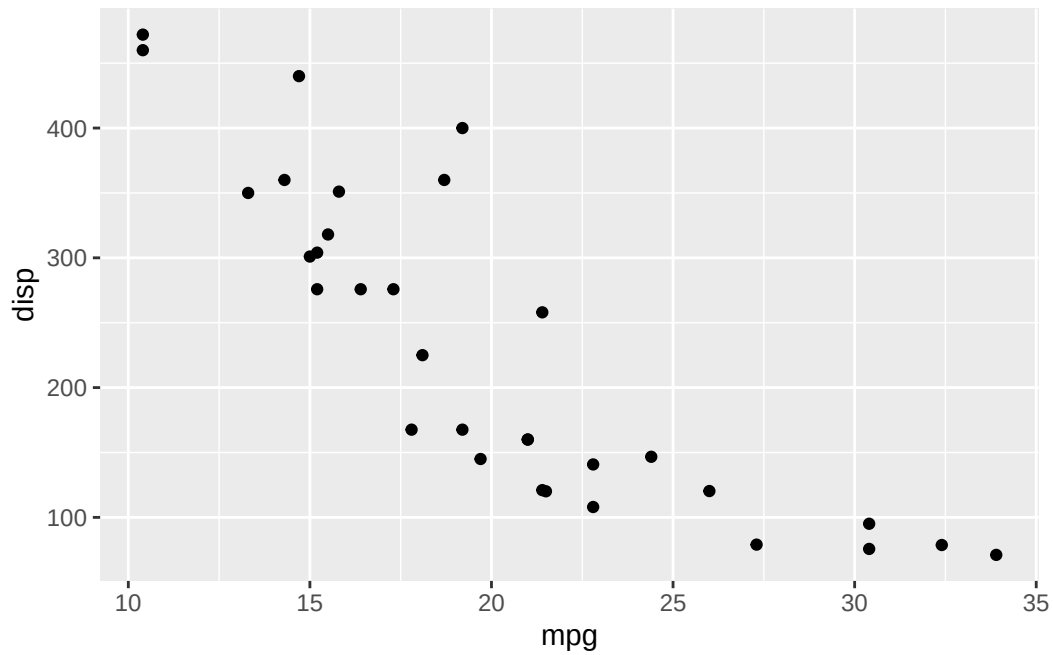
```
# install.packages("ggplot2")
library(ggplot2)
library(colorspace)
library(ggrepel)
library(patchwork)
```

Base R

```
head(mtcars, 5)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2

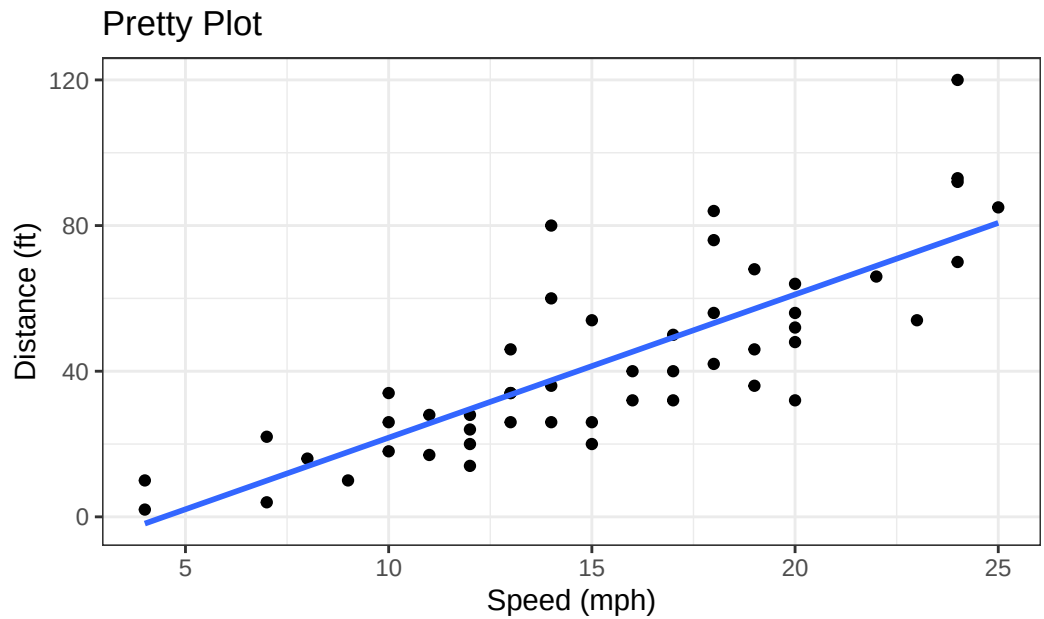
```
ggplot(mtcars) +
  aes(mpg, disp) +
  geom_point()
```



ggplot

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(
    title="Pretty Plot",
    x="Speed (mph)",
    y="Distance (ft)",
    caption="is it really?",
  )
```

``geom_smooth()`` using formula = 'y ~ x'



is it really?

Plot gene expression data

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes, 5)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging

```
ncol(genes)
```

```
[1] 4
```

```
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
table(genes$State)
```

down	unchanging	up
72	4997	127

Q1. How many genes are in this wee dataset?

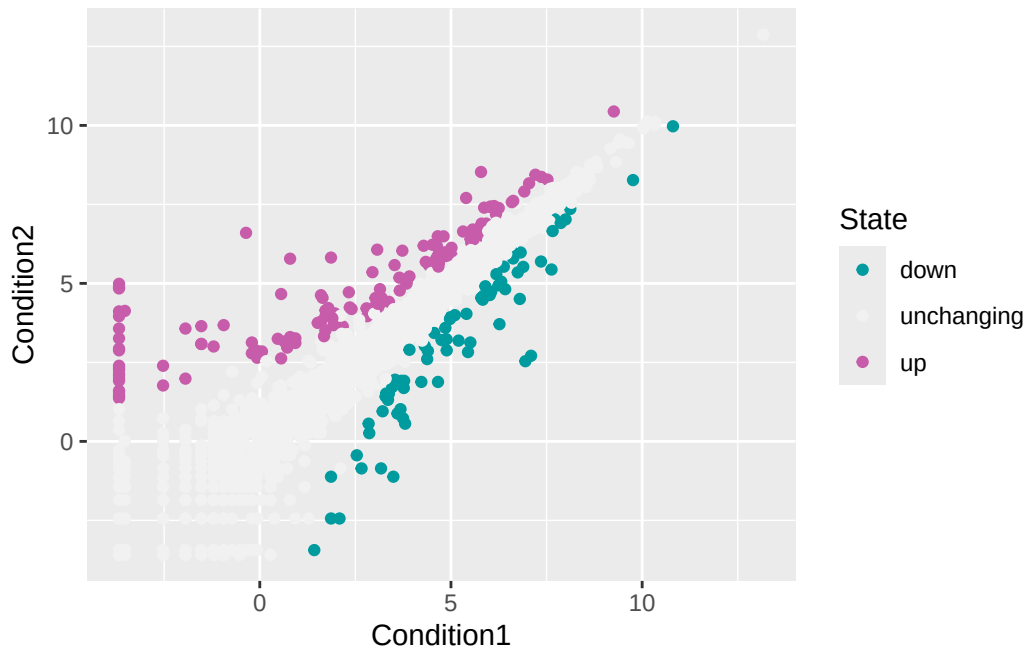
There are 5196 genes in the dataset.

Q2. How many “up” regulated genes are there?

```
# summary of Categorical column
state_df <- table(genes$State)
state_df
```

down	unchanging	up
72	4997	127

```
palette <- diverge_hcl(3, palette="Tropic")
ggplot(genes) +
  aes(
    Condition1,
    Condition2,
    color=State,
  ) +
  geom_point() +
  scale_color_manual(values=palette)
```



```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, color=continent, size=pop) +
  geom_point(alpha=0.2) +
  facet_wrap(~continent)
```

